

SioC 2025: Emerging Technologies and Well-Being

by Lorraine Gaudio

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BOISE STATE UNIVERSITY

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1. Introduction

The second annual workshop for the Social Impacts of Computing (SioC) group was held in August 2025 to engage participants in conversations about how emerging technologies might contribute to human flourishing. Critical to the mission of this event was its inclusion of individuals from a wide range of views in shaping the conversations around the design and use of emerging computing technologies. A key feature of the SioC 2025 workshop was its thematic focus on distinct domains of impact: XR and Educational Technology, Data Ethics, Environmental Impacts, and Healthcare. Across these domains, discussions implicitly invoke various social values – from equity and inclusion in ed-tech, to transparency and responsibility in data practices, to sustainability in environmental technology, to empathy and care in health innovations. This two-day workshop provided 71 individuals from education, small business, government and public sector, healthcare, technology fields, and finance a place to collaborate, learn from, and form partnerships with each other.

2. Program

This workshop was supported by the School for the Digital Future and the School of Computing. Additionally, corporate sponsors were AWS, Pariveda Solutions and Caylent. Additional support came from the College of Innovation and Design and the Division of Research and Economic Development. Attendees were invited through the SioC mailing list and by personal invitations from members of the SioC community. Additionally, 25 participants joined the community showcase, a rolling slide deck that provided a brief highlight of their work, projects, or hobbies related to technology and its impacts on society.

Manuel Gómez-Navarro and Lee Mitchell set the tone with “The Empathic Experience: XR and the Future of Holistic Learning,” exploring how immersive tech can deepen empathy, presence, and perspective-taking while guarding against distraction and overload. The follow-on panel broadened this lens: Mitchell brings data-driven evaluation of immersive tools for student success; Wonda (AI avatar) focuses on human-centered, conversational engagement in XR; and HCI scholar Jerry Alan Fails contributes long-running work on participatory design and the ethics of XR for youth. After a short break, participants split into conversation themes of their choosing (unconferencing). Conversations around how technology might make our lives as workers, friends, thinkers and learners better were summarized by each participant with a key takeaway on sticky notes and whiteboards.

A cross-sector panel examined responsible AI from campus to industry. School of Computing leaders (Ira Burton, Tim Andersen) contextualized institutional strategy and research culture. Pariveda’s Alan Henson added a practitioner’s view from AI/ML and cloud modernization (panel roster and timing). Participants mingled afterward, discussing their topics of interest surrounding data ethics. Popular topics include, “How do we incentivize ethical frameworks in the places needed, such as tech innovation and production?” and “How do we work out what’s healthy to offload to AI?” The afternoon concluded with a keynote from the health-tech founder Alice Crisci, who highlighted “AI with Empathy,” describing how ethically aligned tools can reduce administrative burden and bring clinicians and patients closer together—making care more compassionate and human-centric. The evening concluded with “Happy Hour” where attendees were able to continue their discussions and strengthen the connections they had made throughout the day.

The second day of the workshop was opened with Micron’s Amber Bieg presenting “Sustainability & AI: Micron at the Leading Edge,” surveying sustainability progress and the challenges/opportunities AI introduces for manufacturing and operations. The panel likely connected facility-level energy strategy and technology choices: Bieg’s corporate sustainability work, INL’s Maryl Fisher on building energy analytics and benchmarking for adaptive infrastructure, and ECE’s Omiya Hassan, pointing toward energy-efficient AI hardware and embedded systems—all framed within the session’s focus on AI’s environmental footprint. As the day before, attendees split into unconferencing conversations, themed groups that asked questions like, “Is it better to think about energy use holistically, or on a case-by-case basis?” and “How to think about problems we create?” Key takeaways were recorded on sticky notes and whiteboards.

After lunch, Crisci returned with a hands-on session, “Creating CustomGPTs to Hack your Life, your Health, or your Business,” demonstrating how quickly tailored GPTs can streamline processes and inviting participants to prototype their own tools. The panel then bridged governance, education, and implementation: St. Luke’s Molly Zimmer discussed responsible AI adoption and an in-house Emerging Technologies Lab for real-world testing; Boise State nursing professor Sarah Llewellyn brought perspectives on empathy, simulation, accessibility, and AI integration in clinical education; and Crisci linked product design to patient-centered impact. Following the event, participants mingled and snacked. Many continued the conversations at an off-campus social.

3. Methodology

Data used for this analysis came from three sources. First, the participants pre-registered for the event by filling out a participant registration form, which asked questions regarding their organization associations, what the phrase “social impacts of computing means and its associated values, and up to three collaboration tags. Second, we recorded the content from the anonymous sticky note and whiteboard unconferencing key takeaways. Finally, A post-SioC workshop survey invited participants to review their workshop experience. The survey included a Likert scale evaluation of the workshop activities, as well as asking for ideas for future topics and speakers.

A mixed-methods evaluation of the participant registration form, attendance records, and unconferencing key takeaways investigated the impact of the workshop and the values of our community. First, an analysis of attendance records provides insight into how effectively the pre-event outreach—speaker bios, program details, and reminder emails—converted reservations into in-person attendance, and how many participants were retained from Day 1 to Day 2. We ask, “Among registered participants, what proportion actually attended on each day?” and “Among people who RSVP’d for both days, what share returned for Day 2?” According to the [IT Knowledge Bank](#), event organizers should expect 40 % of individuals who registered to actually attend. Similarly, [Eventbrite](#) reports no-show rates between 30 and 50 %. Based on these benchmarks, attendance success rates are defined as $\geq 70\%$ attendance on Day 1 (Monday) and $\geq 60\%$ attendance on Day 2 (Tuesday). Success is defined as $\geq 80\%$ return rate among participants who RSVP’d for both days.

Second, we ask, “What topics are important to the SioC community?” Using participant selected collaboration tags, themes that emerged from the unconferencing key takeaways, and post-survey ideas, we illustrate emerging topics for future SioC events.

Third, the definitions and associated values of “social impacts of computing” and unconferencing key takeaways reflect the top-of-mind ideas and social values of community members. Values emerge through discourse, culture, and practice (Gunder et al., 2025). By empirically mapping the values the SioC community invokes when they talk about social impacts of computing, conversation advances beyond assumed ideals, toward a grounded understanding of what matters to this community in real-world discussions of technology and society.

Finally, we highlight feedback from the event

4. Results and Discussion

4.1 Attendance

4.1.1 General Attendance

Attendance records are used to gauge how well the topics drew and kept the audience. Of 88 RSVPs, a total of 71 participants attended the SioC workshop, 62 on Monday and 51 on Tuesday (see Table 4.1). 80.5 % of Monday RSVPs actually attended, which comfortably clears the 70 % target. Additionally, Tuesday also exceeded the 60% benchmark by 10.8%. The retention rate was 3.5 % below target. This might be due to the wide range of themes, which may have drawn specific audiences for particular themes.

Table 4.1: Attendance Summary with Targets

Day	RSVP	Actual	Rate	Target
Monday	77	62	0.805	0.70
Tuesday	72	51	0.708	0.60
Either day	88	71	0.807	0.75
Return Rate	61	39	0.639	0.80

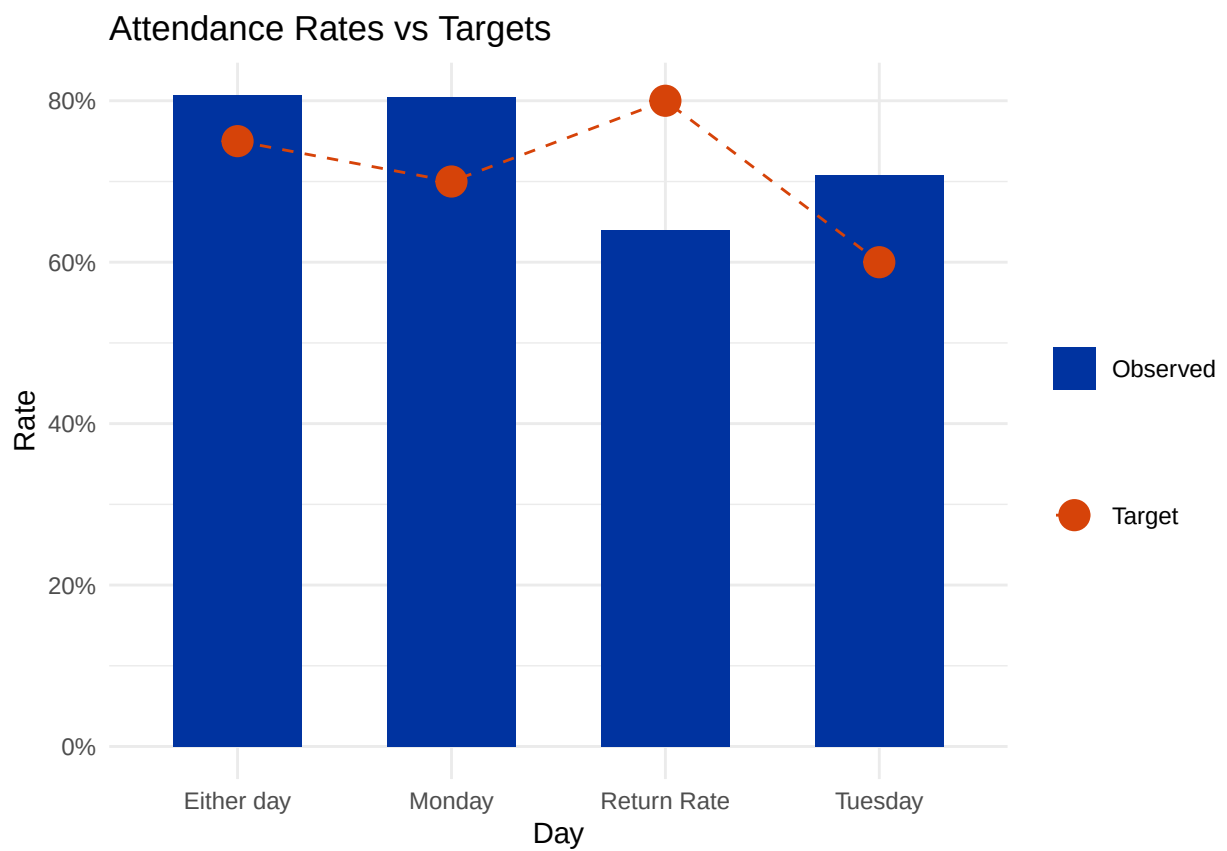
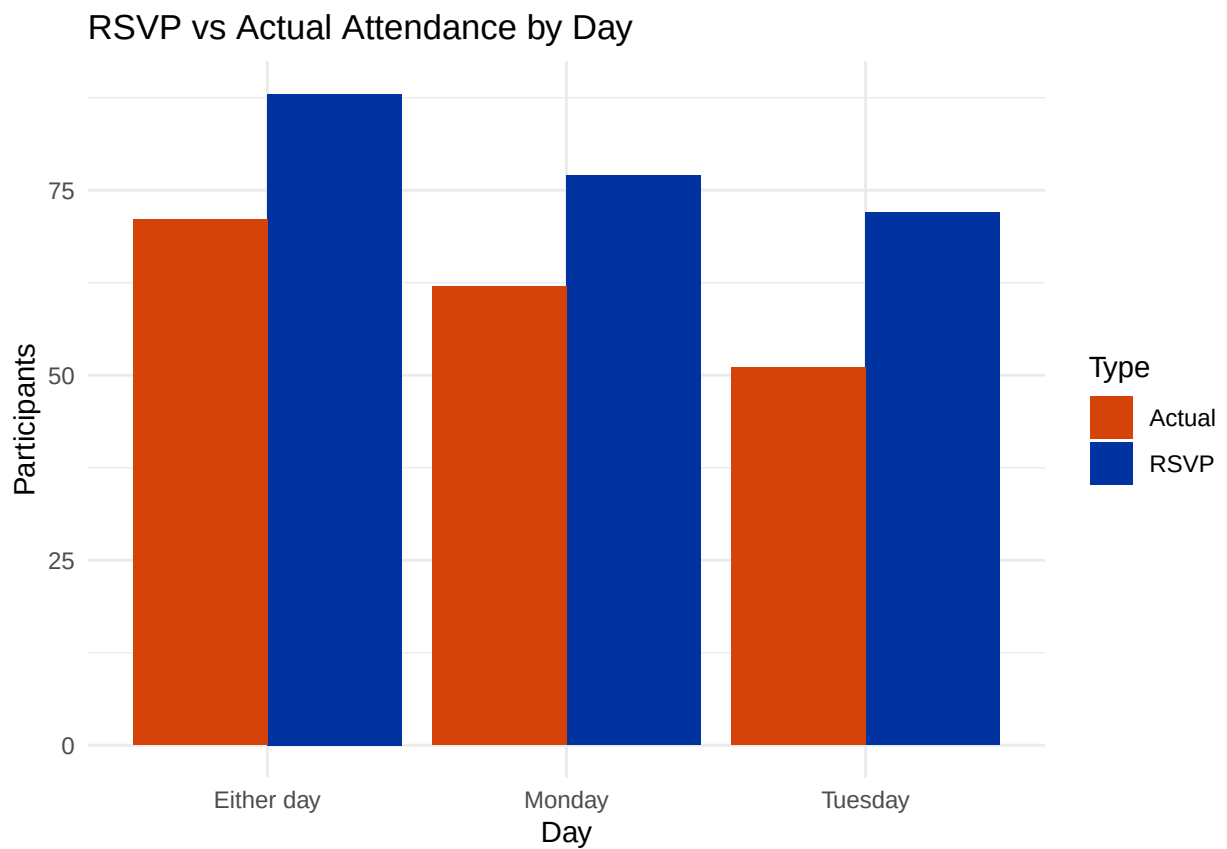


Table 4.2: Attendance by Organization Sector

Sector	Members	Mon	Tue	Either	Rate
Education	48	29	29	35	0.729
Tech & Cloud	13	10	4	11	0.846
Professional Services / Small Business	10	9	6	9	0.900
Government & Public	9	6	5	7	0.778
Health & Healthcare	6	3	4	4	0.667
Individual / Independent	2	2	2	2	1.000

4.1.2 Sector Attendance

We asked registrants to list their organization or institution and grouped responses into compact sectors using regex-based keyword matching: Education, Professional Services / Small Business (including Finance & Insurance), Government & Public, Health & Healthcare, Tech & Cloud, and Individual / Independent. For each sector, we summarized distinct member counts and actual attendance by day, with either-day rates calculated as the proportion of RSVPs who attended.

Table 4.2 shows that Education was the largest sector ($n = 48$), with steady engagement across both days (Either-day rate = 0.812). Professional Services / Small Business, now including finance and insurance organizations, formed the second-largest group ($n = 20$) and demonstrated strong attendance overall (0.800), though turnout on Tuesday was lower than Monday. Tech & Cloud and Health & Healthcare were smaller but showed comparably solid attendance (both 0.800), despite Tech & Cloud's drop-off on Tuesday. Government & Public participation was weaker (0.444), indicating room for improvement in outreach or scheduling alignment for that sector. Individual / Independent participants were few in number but highly engaged, with a perfect Monday showing and an Either-day rate of 1.000. Overall, participation was concentrated in Education and Professional Services, with consistent but smaller contributions from health, tech, and government sectors.

Table 4.3: Attendance by Discipline

Discipline	RSVP	Mon	Tue	Either
Engineering & Technology	16	11	9	13
Humanities	9	8	9	9
Education	7	4	3	4
Business/Economics	6	4	4	4
Natural & Environmental Sciences	3	1	2	2
Healthcare	2	1	2	2

We identified the positions of Boise State faculty and staff who filled out the RSVP form and then manually coded each individual into one of six non-overlapping disciplines: Engineering & Technology, Humanities, Education, Natural & Environmental Sciences, Healthcare, and Business/Economics. For each discipline, RSVP counts distinct faculty/staff who sign up to attend for at least one day; Mon and Tue count the number who actually attended on that day; Either counts

unique people who attended on Monday or Tuesday (including those who attended both days). Rate is a day-2 return rate among both-day RSVPs who started on day-1, computed as the proportion of participants who (i) RSVPed for both days and attended Monday, and (ii) also attended Tuesday.

Table 4.3 shows that Engineering & Technology supplied the largest share of BSU attendees (16 RSVPs; 13 unique across either day), with strong day-1 turnout (11) and a solid return rate (0.812). Humanities was the next-largest cohort (9) and the most consistently engaged—everyone who started returned (Rate = 1.000), with slightly higher participation on day 2 (9 vs. 8), consistent with interest in human-centered XR/ethics and the broader unconference discussions. Education was smaller (7; 4 either-day) with a lower return rate (0.571), suggesting competing commitments during the school term. Business/Economics (6; 4 either-day; Rate = 0.667) showed steady but modest attendance, while Natural & Environmental Sciences (3; 2 either-day; Rate = 0.667) and Healthcare (2; 2 either-day; Rate = 1.000) were small but well targeted: the sustainability/energy focus on day 2 appears to have pulled NES participants, and the health-tech keynote plus clinical-education/implementation panel likely boosted Healthcare’s day-2 presence (2 vs. 1). Overall, the agenda’s blend of XR/human-centered computing, responsible AI, and sustainability resonated most with Engineering & Technology and Humanities, with specific sessions amplifying interest among NES and Healthcare.

4.2 Community Important Topics

A mixed-methods, multi-source pipeline is used to identify community-relevant topics. The collaboration-tag frequencies are quantified, and a thematic analysis on three text corpora: participants’ definitions of “social impacts of computing,” post-survey free-text (most valuable aspects and priority topics), and unconferencing notes. Undoubtedly, artificial intelligence is an important topic for participants. Concern for responsible adoption of AI and data-intensive tech cut across all corpora, but always braided with human flourishing (equity, well-being, agency) and sustainability (energy/water use, ecological impacts). The unconferencing key takeaway analysis showed that XR/EdTech contributions emphasized human connection and authenticity in learning contexts where as the Environment sessions stressed prioritization and public buy-in for sustainable AI.

4.2.1 Collaboration Tags

Figure @ref{fig:collab-tag} summarizes collaboration interests using participants’ multi-select tag choices (up to three choices each). After excluding “NA (no answer),” we analyze 246 total tag selections. To keep the figure readable, topics representing <3.9% of selections are grouped into Other. The plurality tag is Artificial Intelligence (24.0%), followed by Innovation (10.6%), Technology Adoption (9.3%), Ethics (8.5%), Education (8.5%), Culture (7.3%), and Business & Economics (6.5%); additional notable tags include Environment (5.3%), Data Security (5.3%), XR (Extended Reality) (4.1%), Healthcare (4.1%), and Digital Equity (4.1%). Other (2.4%) aggregates low-frequency tags—Legal (n=3), Faculty Development (n=1), Research (n=1), and Human Frailty (n=1).

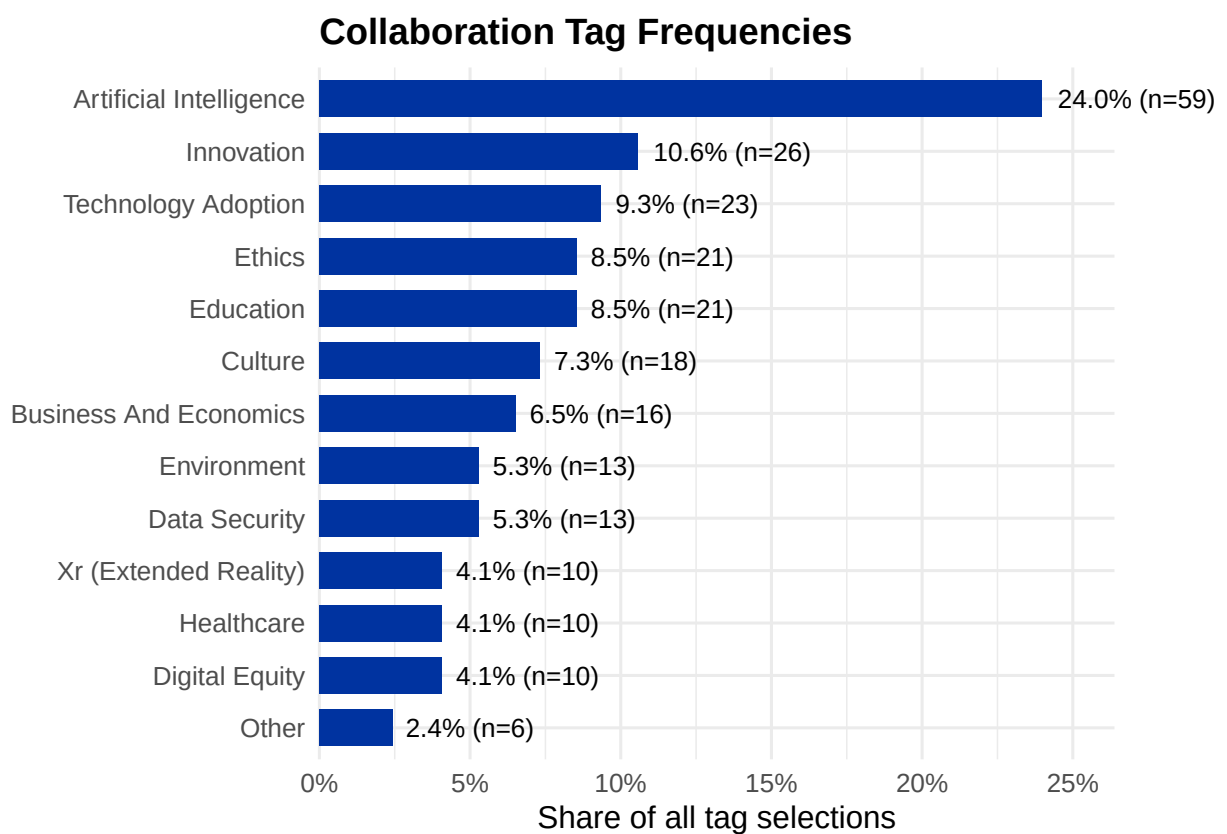


Figure 4.1: Collaboration Tag Frequencies. Share of 246 tag selections (multi-select; 'NA' removed). Tags <3.9% are grouped into Other (Legal, Faculty Development, Research, Human Frailty).

4.2.2 Three Text Corpora

Three text corpora (participants' definitions of "social impacts of computing," post-event free-text, and notes from XR and Environment unconferencing) were analyzed to identify the topics most on people's minds and the values that animate them.

4.2.2.1 Topics

A single shared topic dictionary—seeded from the workshop tracks (e.g., XR/EdTech, Data Ethics/Governance, Environment, Healthcare) plus cross-cutting areas (AI adoption/workforce, equity/access, privacy/security, business/economy, culture/education) were discovered manually through thematic analysis coding. For each topic, source-specific frequencies and a composite "importance score" were produced (See Table 4.5).

Table 4.4: SioC Topic Domains and Abbreviations

Topic	Code	Definition
XR/EdTech	XR	Immersive/interactive technologies (VR/AR/MR) and digital learning tools; pedagogy and student success, accessibility and safety, cognitive load, and classroom integration.
Data Ethics/Governance	DE/G	Rules, norms, and practices for responsible data/AI: fairness & bias, accountability, provenance, transparency, consent/ownership, and governance frameworks.
Environment	ENV	Environmental impacts across the compute lifecycle (energy, emissions, water, materials, e-waste) and the use of tech/models for sustainability and environmental justice.
Healthcare	HLTH	Computing in health and care: diagnostics and decision support, digital therapeutics, privacy/safety (e.g., HIPAA), equity, and accountability in clinical contexts.
AI adoption/workforce	AI/WF	How organizations and individuals adopt AI; change management, training/upskilling, job redesign, productivity impacts, and responsible deployment practices
Equity/access	EQUITY	Fair access to technology and its benefits; accessibility/ADA, affordability, inclusive design, and bridging the digital divide for marginalized communities.
Privacy/security	PRIV	Protection of personal/sensitive data and systems: cybersecurity, consent, anonymization/de-identification, safe data sharing, and incident response.
Business/economy	BUS	Economic uses and impacts of computing: entrepreneurship, ROI and business models, regulation/compliance, and local innovation ecosystems.
Culture/education	CUL/EDU	How computing shapes culture, communication, and learning; pedagogy and assessment, academic integrity, media/digital literacy, and civic learning.
Well-being/Human Flourishing	WBF	Effects of computing on mental/physical well-being and dignity: autonomy, safety, ergonomics, time/attention, pro-social design, and community resilience.

Table 4.5: Topic Importance (Within-Source Shares and Average Importance)

Topic	Share: Define	Share: Unconf	Share: Valuable	Share: Priority	Importance
CUL/EDU	31.9%	16.8%	61.1%	21.7%	32.9%
DE/G	17.7%	24.8%	11.1%	17.4%	17.7%
WBF	20.6%	14.9%	22.2%	8.7%	16.6%
AI/WF	3.5%	12.9%	0.0%	17.4%	8.5%
ENV	7.8%	7.9%	0.0%	13.0%	7.2%
BUS	7.1%	3.0%	0.0%	13.0%	5.8%
HLTH	1.4%	1.0%	5.6%	8.7%	4.2%
XR	0.0%	12.9%	0.0%	0.0%	3.2%
EQUITY	7.1%	2.0%	0.0%	0.0%	2.3%
PRIV	2.8%	4.0%	0.0%	0.0%	1.7%

4.2.2.2 Values

A SioC community valued dictionary was created using established scholarship on research integrity and values (Resnik and Elliott (2023); Matthewson (2011); Kisselburgh and Beever (2022); Beever and Pinkert (2018)) and then redefined those terms so they better fit SioC's broader civic context (beyond academic science). An **initial code book** was narrowed into nine distinct domains (see Table 4.6).

Table 4.6: Nine Domains of SioC Values

Value	Code	Definition
Adaptability & Skepticism	ADA	Stay flexible and update beliefs/strategies as evidence and conditions change; be willing to be wrong.
Autonomy & Voice	AUT	Protect freedom of inquiry and individual/collective agency; ensure affected stakeholders have a say and can opt-in/opt-out.
Civic Equity	CIV	Act for the public good and ensure fair access, non-discrimination, and due process across groups affected by technology.
Curiosity & Rigor	CUR	Be inquisitive and evidence-driven; value critical thinking, objectivity, explainability, and perseverance in learning.
Economic Development	STE	Use time, money, energy, and data wisely to create value for people and communities; includes continuous improvement and responsible growth.
Privacy & Confidentiality	PRI	Safeguard sensitive information, uphold confidentiality obligations, and minimize, de-identify data.
Respect & Collaboration	RES	Treat people with dignity; share credit; work cooperatively and collegially across roles and sectors.
Responsibility & Accountability	ACC	Anticipate/mitigate harms, own the outcomes, and self-correct when needed.
Transparency & Integrity	TRA	Be open and honest about methods, data, interests, and limitations; share materials appropriately and give fair credit.

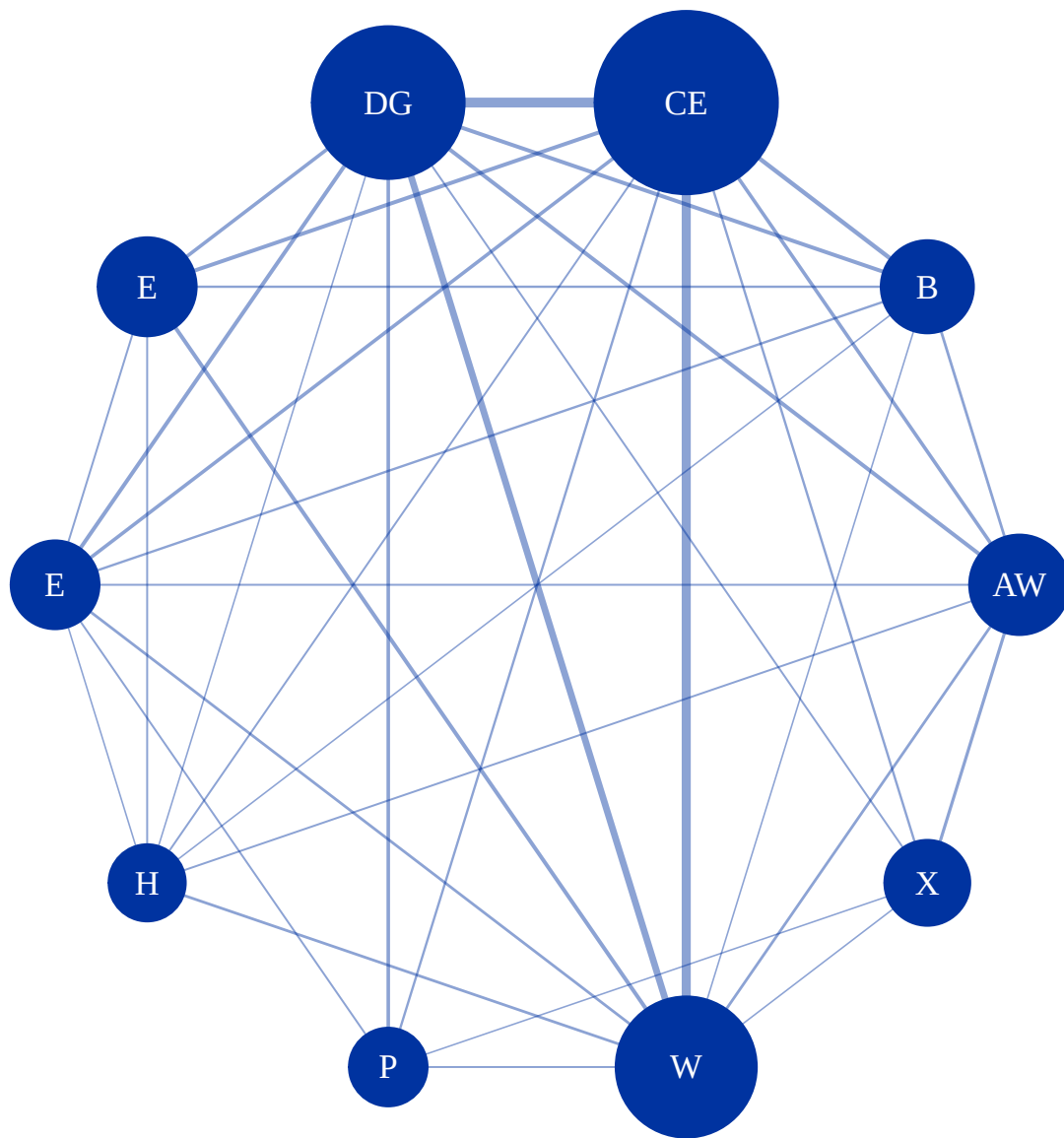
Table 4.7: Top Values Across Sources

Value	Definition	Unconf: Env	Unconf: XR	Post: Valu- able	Post: Make it better	Post: Prior- ity	Total	Share of All
Civic Equity	58	10	1	1	4	7	81	23.1%
Responsibility & Accountability	33	8	8	0	2	5	56	16.0%
Respect & Collaboration	5	6	14	14	8	3	50	14.2%
Curiosity & Rigor	11	12	4	11	2	6	46	13.1%
Economic Development	13	12	6	0	7	5	43	12.3%
Adaptability & Skepticism	10	10	8	2	1	0	31	8.8%
Autonomy & Voice	3	6	7	2	2	0	20	5.7%
Transparency & Integrity	4	5	6	0	1	0	16	4.6%
Privacy & Confidentiality	5	0	3	0	0	0	8	2.3%

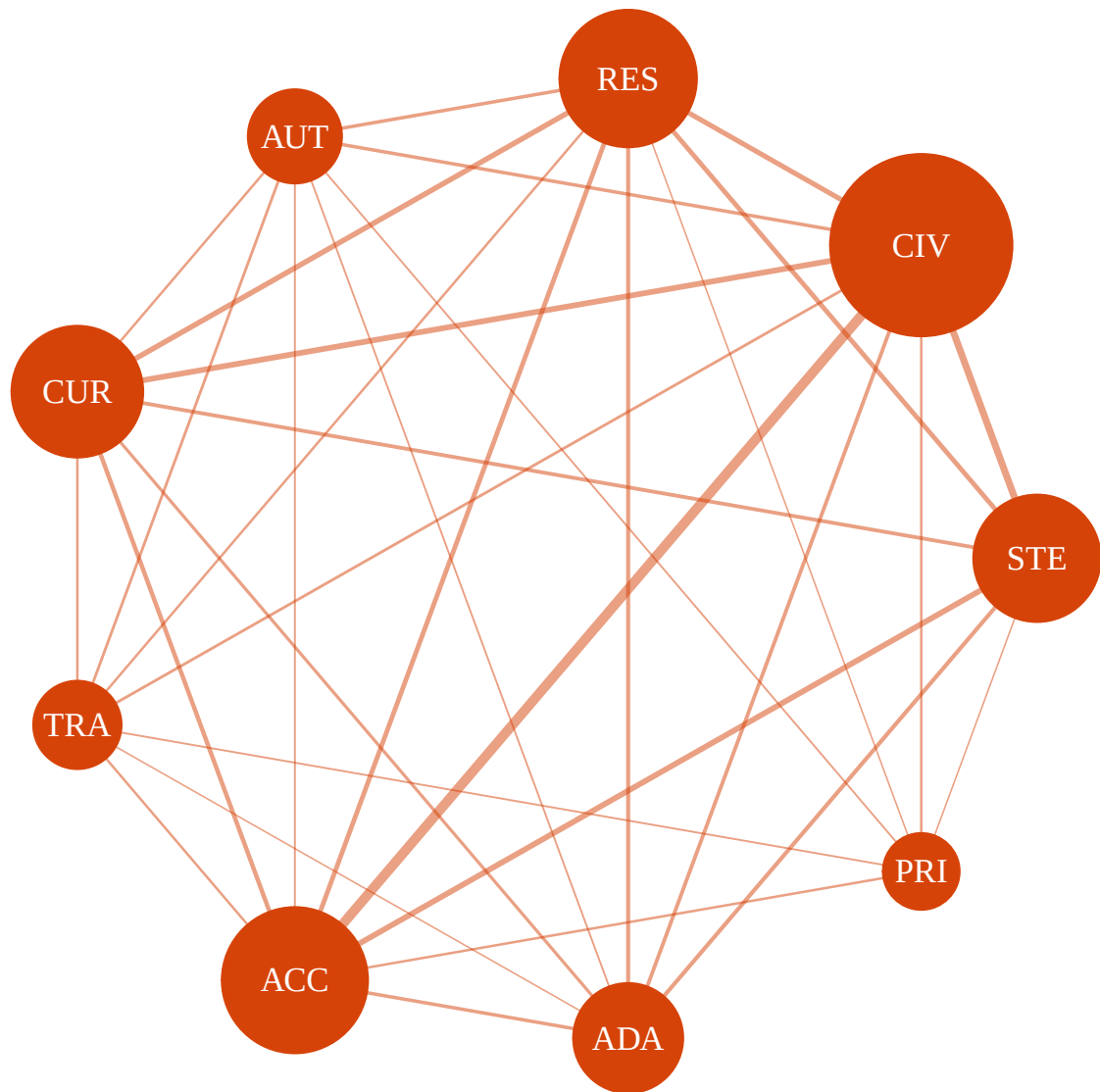
4.3 Topics and Values Mapping

Using ‘igraph’ package (Csardi et al., 2024), a network visualization shows circular layouts to spatially position topics values.

Topic–Topic Co-occurrence (circle layout)



Value–Value Co-occurrence (circle layout)



4.4 Event Feedback

5. Recommendations and Conclusion

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